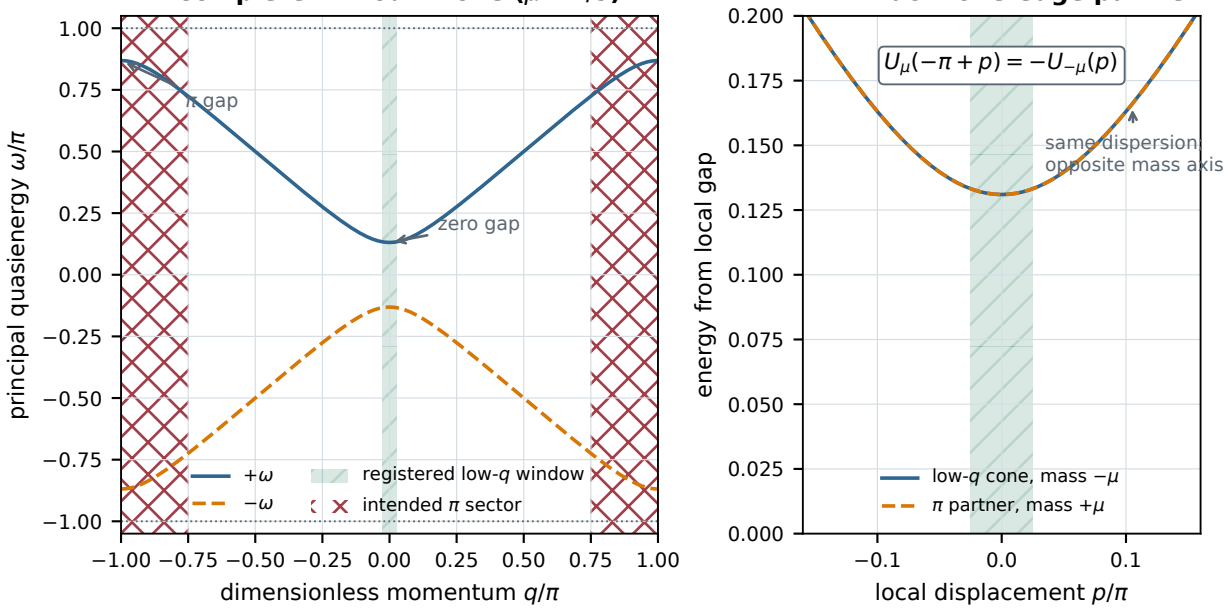
D1 Complete Brillouin zone ($\mu = 2/5$)
principal quasienergy ω/π
dimensionless momentum q/π
π gap
zero gap
$+\omega$
$-\omega$
registered low-q window
intended π sector
D2 Exact zone-edge partner
energy from local gap
local displacement p/π
$U_\mu(-\pi + p) = -U_{-\mu}(p)$
same dispersion,
opposite mass axis
low-q cone, mass $-\mu$
π partner, mass $+\mu$